Team No .60 Report

-This project was implemented by Team No. 60, whose members are:

1- Mustafa Younis (Team Leader) ID: 23011550
2- Abdelrahman Gaber ID: 23011312
3- Ezzat Abdel Ghaffar ID: 23011360
4- kirolos Mounir ID: 23011424
5- Abdelrahman Mohammed ID: 23012002
6- Mina Tharwat ID: 23011576

What this data set contains?

A	B	C	D	E	F	G	H
items	count	total	rnd	customer	age	city	paymentType
citrus fruit	4	1612	9	Maged	60	Hurghada	Cash
tropical fr	3	509	12	Eman	23	Aswan	Cash
whole mil	1	2084	8	Rania	37	Dakahlia	Cash
pip fruit,y	4	788	8	Rania	37	Dakahlia	Cash
other veg	4	1182	14	Magdy	36	Sohag	Cash
whole mil	5	1771	3	Ahmed	30	Giza	Credit
rolls/buns	1	2196	7	Huda	39	Gharbia	Cash
other veg	5	1657	6	Walaa	29	Cairo	Cash
pot plants	1	2373	2	Mohamed	25	Alexandri	Credit
whole mil	2	343	5	Shimaa	55	Port Said	Cash

−We have a dataset consisting of 9836 rows and 8 columns (variables),Each row represents a specific transaction Its details are explained throughout the eight columns(Names of the products that the customer purchased (items) , Number of these products (count) , The total cost of the purchase (total), Each customer's identification

number (rnd) , The name of each client (customer), Client's age(age), The city where each customer lives(city) , The payment method for each customer is either cash or credit(payment type)).

Where is the problem?

−The problem is that these data set may be unformatted, which means there may be duplicated rows or NA Values or Outliers. We also need to visualize this data and represent it in statistical forms to make it easier to read, analyze, and extract important information from. We also need to apply K-Means To know the division of numeric data and how close each element is to the other , On the other hand, we need to apply Association Rule, including in order to know which products are most accepted by customers and which have a high purchase rate, to exploit this point and place them next to each other, or to exploit this in making offers and discounts in .

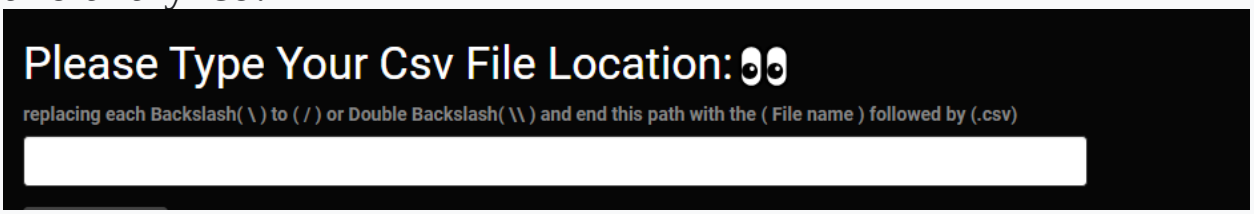
What Will the Program do?

−Here comes the role of our team, we designed our program, The mission of this program is to read the entire file and clean this data by deleting all duplicated rows, deleting any (outliers), deleting any (NA) values, then displays an analysis of this data in a statistical manner, Then applying K-means to separate and divide the digital data, and then apply Association rule by using

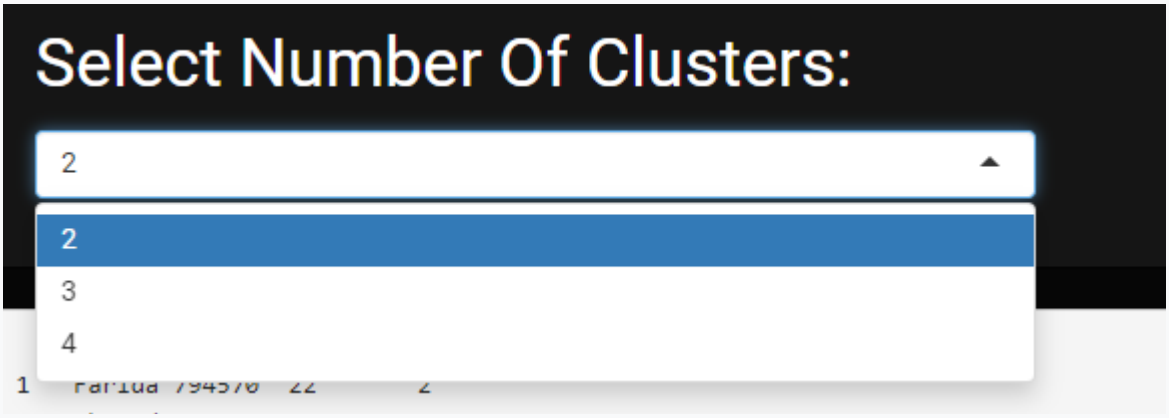
(Apriori Algorithm), As we mentioned, it shows me the most purchased products via Support and confidence.

What the input to the program will be?

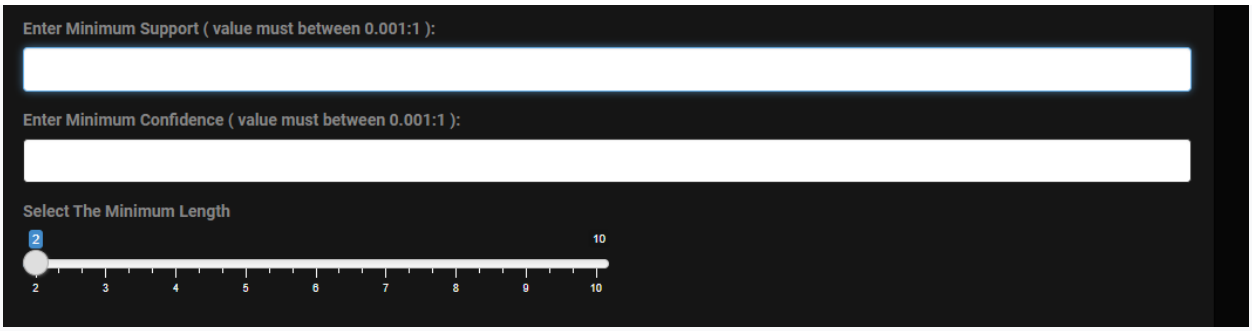
-in the first window ,The required input for the program is the path to the CSV file replacing each Backslash(\ ) to (/) or Double Backslash(\\ ) and end this path with the (File name) followed by (.csv), this file containing a set of data to be cleaned and analyzed.



- Also, in applying k-means window, the user is also asked to choose the number of clusters.



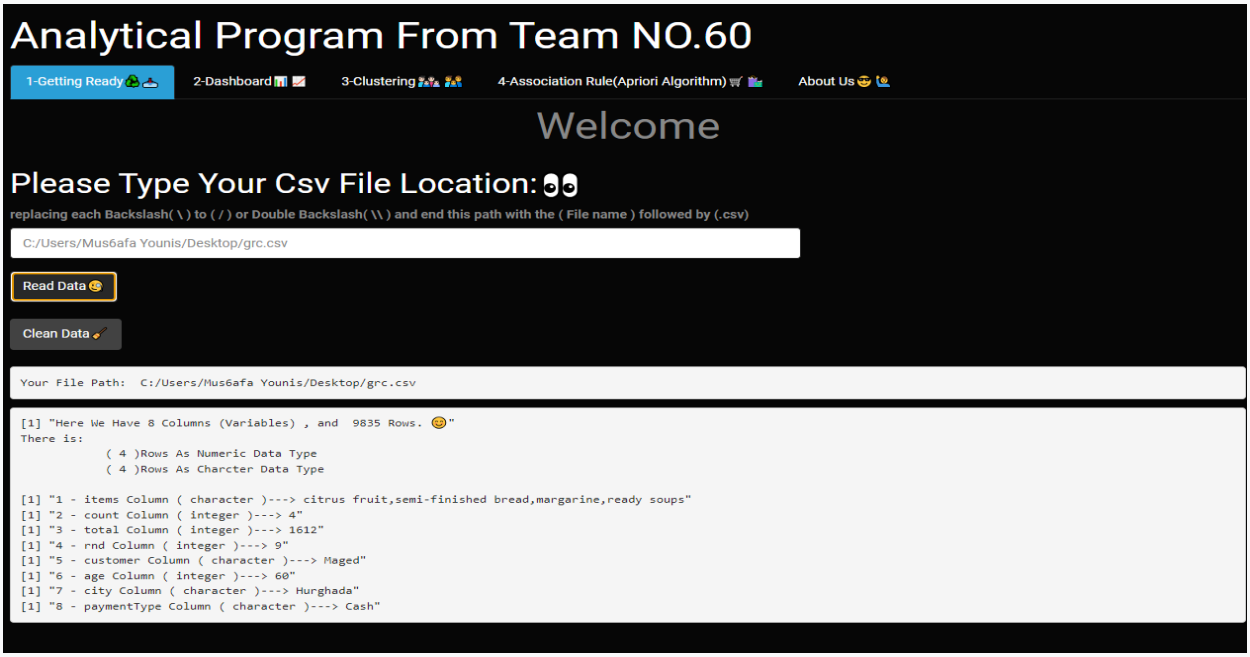
- In the Association rule window in which the Apriori Algorithm is applied, the user is required to set specific and appropriate values for support, confidence, and min length.



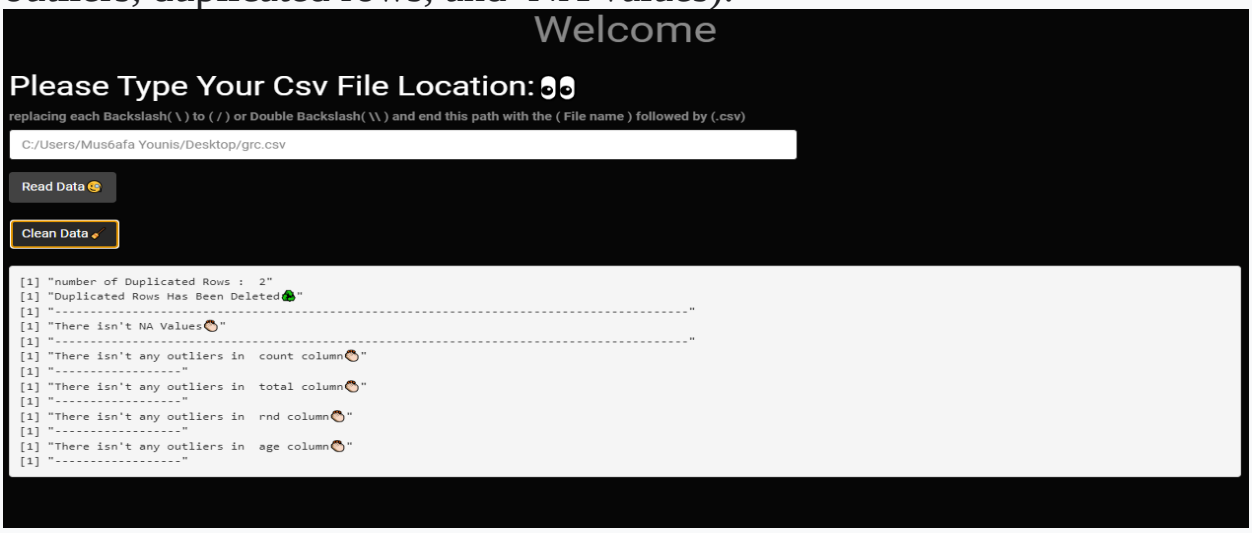
What the output from the program will be?

The output that will come out of the program will be as follows :

- When the Read Data button is pressed, File Path appear a complete explanation of the data set contained in this file appears, for which the user wrote the path to his device in the required manner. It tells him the number of columns and the number of rows, and tells him the type of data for each column of this data set.



– When user press the Clear Data button , A Data Cleaning process is applied such that if duplicate rows are found, these duplicated rows are deleted, and if Outliers are found, the rows containing these are deleted, and if they are found, the rows containing these empty values are deleted (Before applying the deleting process, the user is shown the number of outliers, duplicated rows, and NA values).



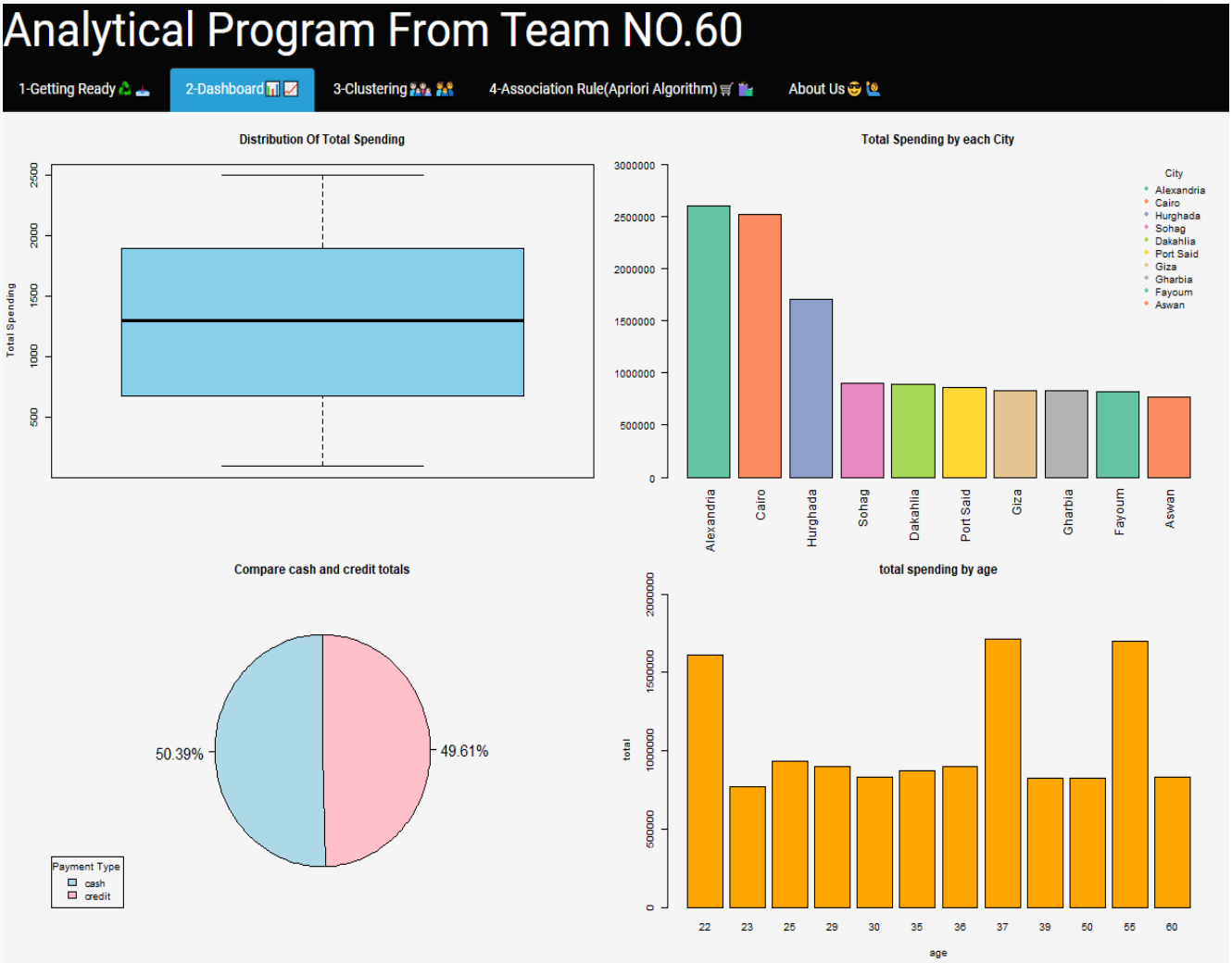
- When you select the Dashboard window An analytical presentation appears in four graphical forms:

1- A barplot comparing age with the total spending of each age.

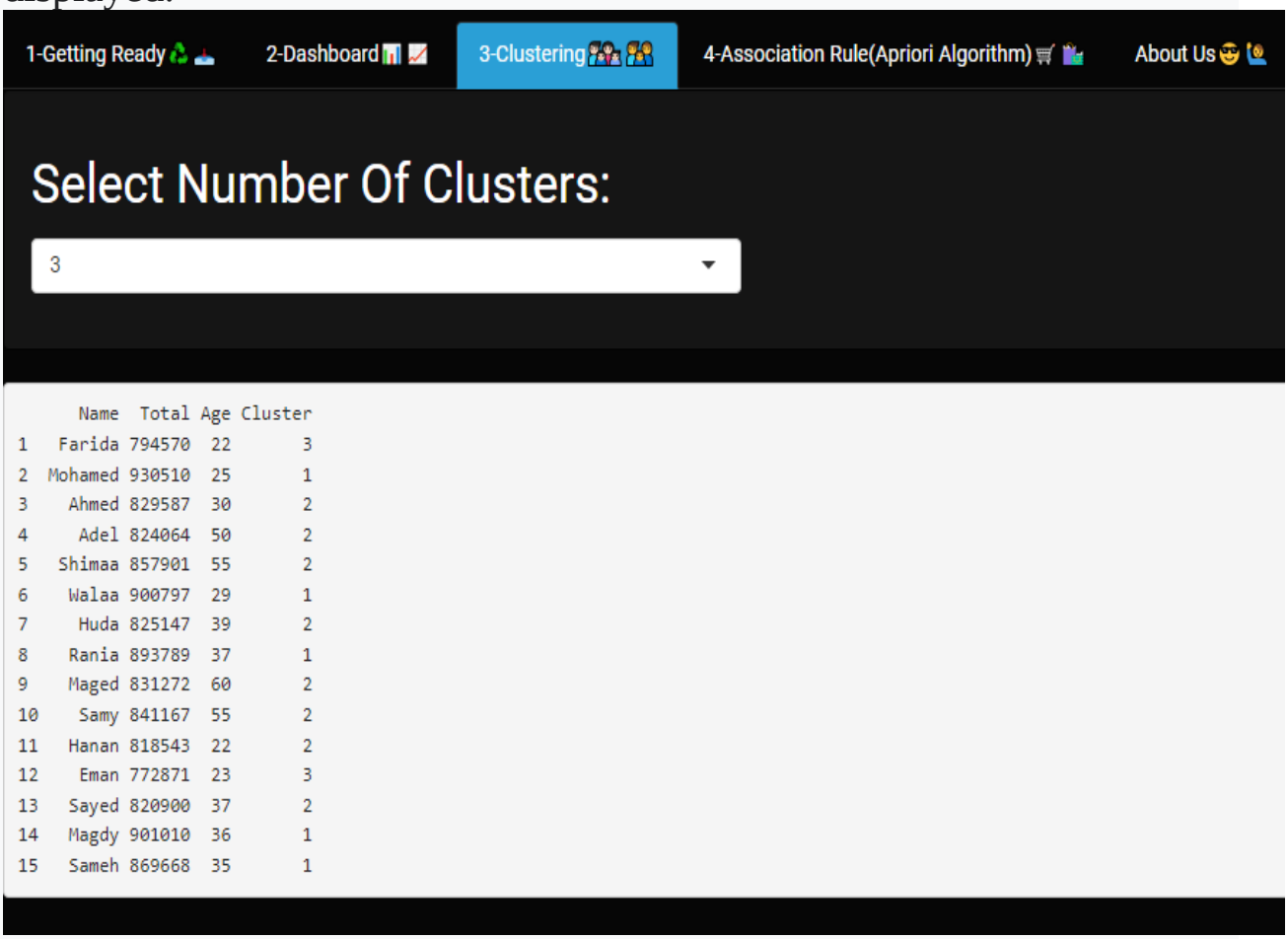
2- A Pie Chart Shows a comparison of payment type between cash and credit.

3- Another bar Plot for making comparison between each city and its total spending.

4- Box Plot represents Total Spending distribution.
Their role in general is an analytical and detailed presentation of the data To make it easier for the user to read and display his data statistically.



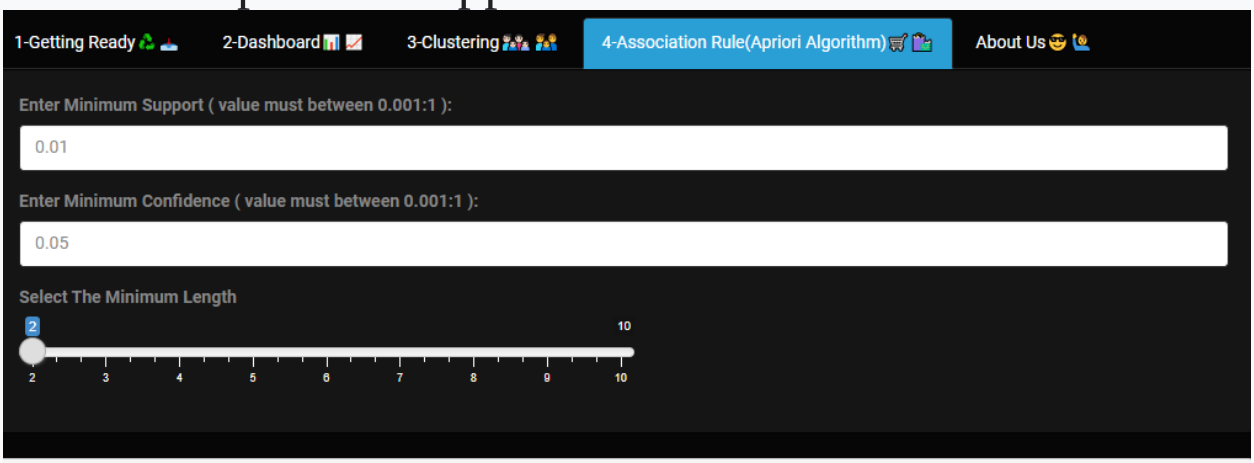
-When the user choose the clustering window and choose the number of clusters between 2 and 4 based on which he wants to divide , the name and age of each customer and his total spending throughout the entire data set are shown to him, and the cluster he has been classified into is displayed.



	Name	Total	Age	Cluster
1	Farida	794570	22	3
2	Mohamed	930510	25	1
3	Ahmed	829587	30	2
4	Adel	824064	50	2
5	Shimaa	857901	55	2
6	Walaa	900797	29	1
7	Huda	825147	39	2
8	Rania	893789	37	1
9	Maged	831272	60	2
10	Samy	841167	55	2
11	Hanan	818543	22	2
12	Eman	772871	23	3
13	Sayed	820900	37	2
14	Magdy	901010	36	1
15	Sameh	869668	35	1

-When moving to the Association rule window, and the user selects minimum Support and minimum confidence,

which are applied to the items in Data set , The apriori algorithm is applied and shows the products that achieve support and confidence with values equal to or higher than the values entered by the user, and based on the value of the min length chosen from the slider, the number of products appears.




```
[1] "Right Values,Here We Go: "
set of 513 rules
      lhs                                   rhs                      support    confidence coverage   lift      count
[1]   {hard cheese}                      => {whole milk}             0.01006609 0.41078838 0.02450432 1.6076815  99
[2]   {butter milk}                      => {other vegetables}       0.01037112 0.37090909 0.02796136 1.9169159 102
[3]   {other vegetables}                 => {butter milk}            0.01037112 0.05359958 0.19349263 1.9169159 102
[4]   {butter milk}                      => {whole milk}             0.01159126 0.41454545 0.02796136 1.6223854 114
[5]   {ham}                              => {whole milk}             0.01148958 0.44140625 0.02602949 1.7275091 113
[6]   {sliced cheese}                    => {whole milk}             0.01077783 0.43983402 0.02450432 1.7213560 106
[7]   {oil}                              => {whole milk}             0.01128622 0.40217391 0.02806304 1.5739675 111
[8]   {onions}                           => {other vegetables}       0.01423488 0.45901639 0.03101169 2.3722681 140
[9]   {other vegetables}                 => {onions}                 0.01423488 0.07356805 0.19349263 2.3722681 140
[10]  {onions}                           => {whole milk}             0.01209964 0.39016393 0.03101169 1.5269647 119
[11]  {berries}                          => {yogurt}                 0.01057448 0.31804281 0.03324860 2.2798477 104
[12]  {yogurt}                           => {berries}                0.01057448 0.07580175 0.13950178 2.2798477 104
[13]  {berries}                          => {other vegetables}       0.01026945 0.30886850 0.03324860 1.5962805 101
[14]  {other vegetables}                 => {berries}                0.01026945 0.05307409 0.19349263 1.5962805 101
[15]  {berries}                          => {whole milk}             0.01179461 0.35474006 0.03324860 1.3883281 116
[16]  {hamburger meat}                   => {other vegetables}       0.01382816 0.41590214 0.03324860 2.1494470 136
[17]  {other vegetables}                 => {hamburger meat}         0.01382816 0.07146611 0.19349263 2.1494470 136
[18]  {hamburger meat}                   => {whole milk}             0.01474326 0.44342508 0.03324860 1.7354101 145
[19]  {whole milk}                       => {hamburger meat}         0.01474326 0.05769996 0.25551601 1.7354101 145
[20]  {hygiene articles}                 => {whole milk}             0.01281139 0.38888889 0.03294357 1.5219746 126
[21]  {whole milk}                       => {hygiene articles}       0.01281139 0.05013928 0.25551601 1.5219746 126
[22]  {salty snack}                      => {other vegetables}       0.01077783 0.28494624 0.03782410 1.4726465 106
```

What is the role of each member?

-as each member of us played an important, essential and effective role in this process to solve these problems, so, In which each person played a specific role As follows:

1- Mustafa Younis : As he is the person responsible for designing the User Interface (UI) using R Shiny Package, and configuring the three basic structures (ui, server, shinyApp), Where The User Interface Function is the specialized function for implementing and displaying graphical interfaces to the user and detailing their shapes and titles, On the other hand, the server's function is concerned with the mathematical and mental operations of the program. It receives inputs from the UI function, exists outputs from it, and sends them to the UI function, Finally, the Shiny App Function comes into play, which is responsible for linking the user interface and the server, and is mainly specialized in running the program.

2-Mina Tharwat : He is responsible for Data cleaning, where his role is that after reading and knowing the data and its circumstances, he does the following: delete duplicated rows, delete empty values(NA values), and delete outliers, that is, values that are abnormal and not in the normal range, which mainly affects the median.

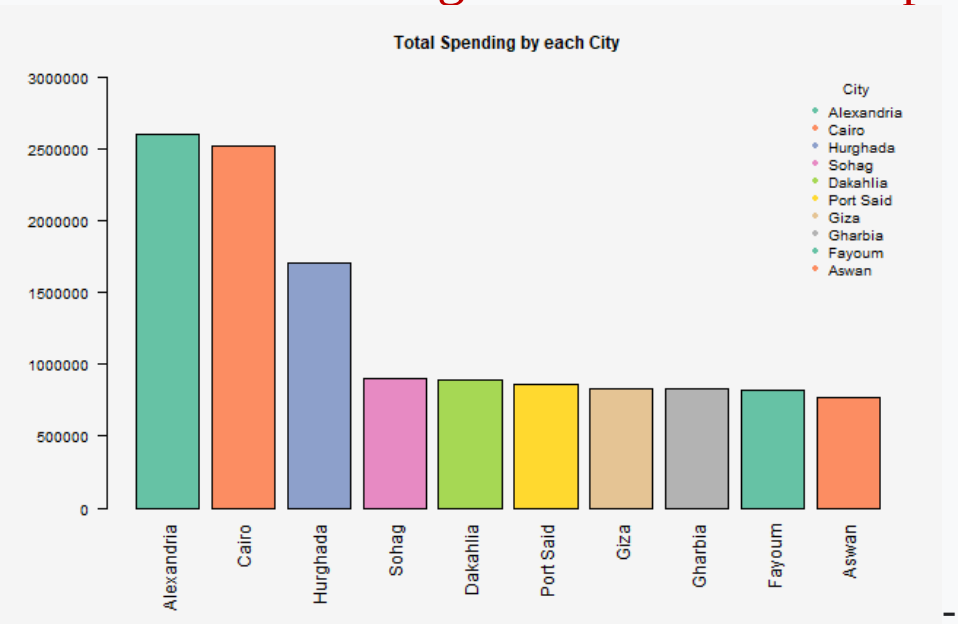
3- kirolos Mounir & Ezzat Abdel Ghaffar : Their role lies in that they are responsible for the Data Visualization, that is, making an analytical presentation of the data in a statistical and organized manner to make it easier for the user to read. They represented four graphical forms:
First: the bar plot, which represents a comparison between age and total spending.
Second: the box plot, which represents the complete distribution of the total spending.
Third: another barplot for comparing each city with its corresponding total spending.
Finally: they represented the pie chart for illustrating a comparison between payment methods, namely cash and credit.

4- Abdulrahman Gaber : As he played an effective role in representing K-means based on the number of clusters chosen by the user, he filtered the entire data from the beginning, as he collected the Total Spending for each client separately without repetition and represented each customer name, age, total spending, and cluster to which it belongs to , in the form of 15 clients, that is, 15 row, so each person has the cluster to which he belongs.

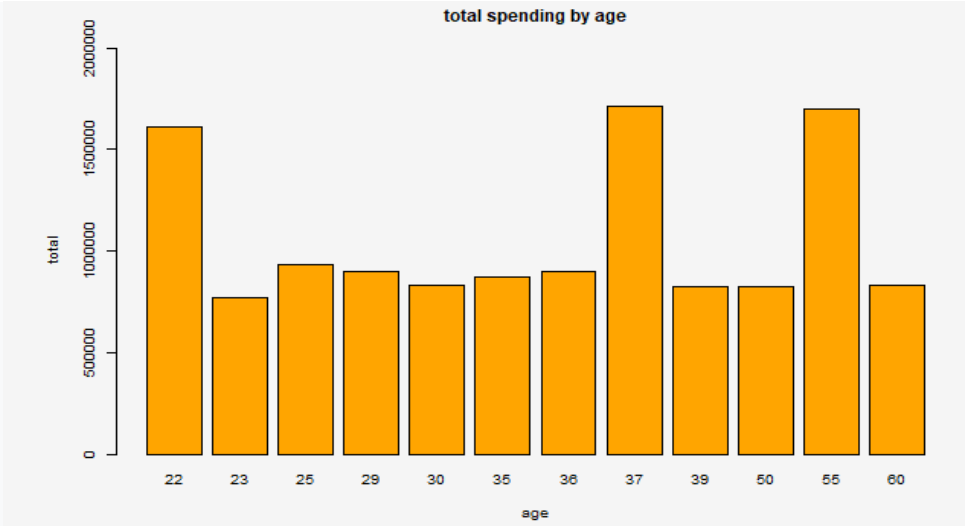
5- Abdulrahman Mohammed : He was responsible for implementing the Association rule and using the Apriori

Algorithm, and his role in general is that he receives from the user the value of support and confidence, and based on these values, he represents the most purchased and most popular product groups based on the percentage of support and confidence and the number of products based on the value of min length.

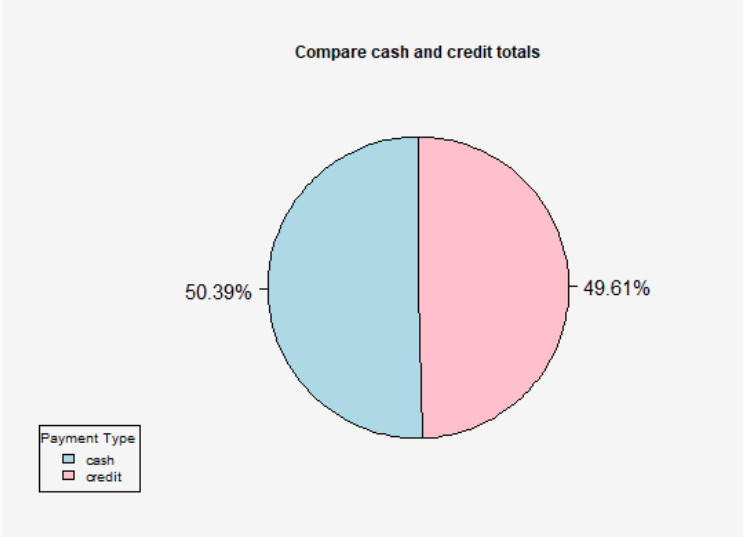
Our results and insight from Plotted Graphs:



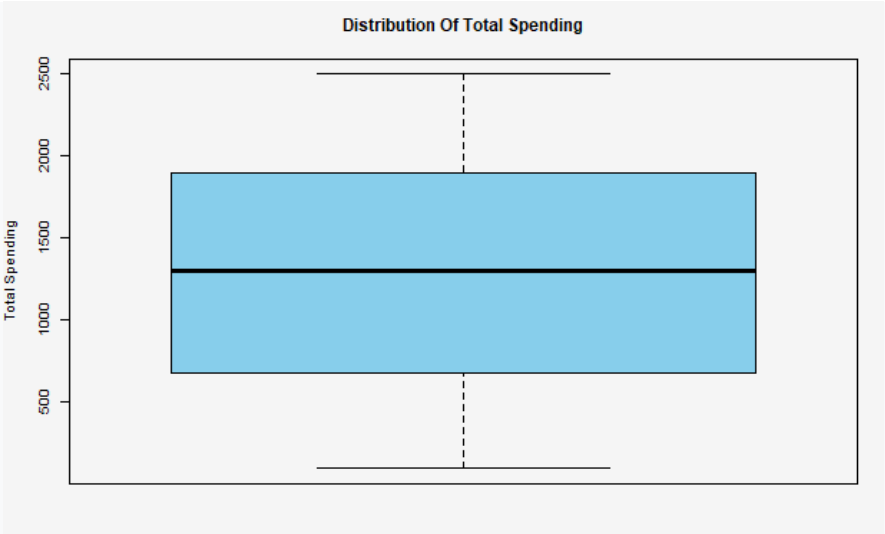
- As we see here in the barplot for comparing each city with its total spending, we find that the city with the most consumption and the largest purchasing value(Total Spending) is Alexandria, while the corresponding city with the lowest total spending is Aswan.



-As we see here in the barplot for comparing each age group with its Total Spending, we find that the most consuming age group is the 37 and 55 years, while the least consuming age group has the lowest Total Spending, which is the 23 years.



- As we see here in the pie chart for comparing payment methods between cash and credit, we find that there is a closeness between them, but the percentage of payment methods using cash is higher than credit by a small percentage.



–finally, In the box plot distributing the values of total spending per customer, we find that the minimum value of total spending is 100, the maximum value is 2500, the first quartile is 679, the third quartile is 1896.5, and the Median is 1297.

-Explanation of the code :

```
#Libraries that were used

library("shiny")
library("shinythemes")
library("reader")
library("magrittr")
library("dplyr")
library("RColorBrewer")
library("plotly")
library("arules")
```

-Libraries that were used:

-library("shiny") : **This library enables you to build interactive web applications using the R programming language. It allows users to interact with data and analyses dynamically.**

-library("shinythemes"): **This library provides a collection of ready-made themes to decorate your Shiny applications, giving you more options to customize the look and design of your application.**

-library("reader") : **This library helps in reading data files from various sources such as CSV, Excel, and others. It makes reading data easier and more flexible.**

-library("magrittr") : **This library is used to make your R code more readable and understandable using the pipe operator %>%. These pipes are used to sequence functions and make the code less complex.**

-library("dplyr") : **This library provides a powerful set of tools for data manipulation and transformation. It allows you to perform**

common operations such as filtering, summarizing, and comparing data easily and efficiently.

-library("RColorBrewer") : **This library offers a wide range of color palettes designed specifically for use in graphical visualizations. It enables you to choose suitable colors for your data and enhance the end user experience.**

-library("plotly") : **This library is a powerful tool for creating interactive graphs in R. It allows users to interact with the graphs, such as zooming and panning, making it easier to understand and explore data.**

-library("arules"): **This library provides tools for data analysis using association rule mining techniques. It is typically used in analyzing behaviors and trends in data such as market basket analysis and others.**

-Explanation of the UI :

```
#Shiny_User_InterFace

ui <- fluidPage(

  theme =shinytheme("cyborg"),

  titlePanel("Analytical Program From Team NO.60"),

  tabsetPanel(
    tabPanel("1-Getting Ready",
             div(style = "text-align: center; font-size: 45px;","Welcome"),
             h3("Please Type Your Csv File Location:\n"),
             textInput("Data_Path","replacing each Backslash( \\ ) to ( / ) or Double Backslash( \\\\ ) and end this path with the ( File name ) followed by (
             actionButton("Read_Button","Read Data"),
             h1(""),
             actionButton("Clean_Button","Clean Data"),
             h1(""),
             verbatimTextOutput("Data_Path"),
             verbatimTextOutput("Data_Str"),
             verbatimTextOutput("Data_Clean")),
    tabPanel("2-Dashboard",
             plotOutput("Dashboard",width = "1250", height = "810")),
    tabPanel("3-Clustering",
              sidebarLayout(
               sidebarPanel(selectInput("Number_Of_Clusters",h3("Select Number Of Clusters:"),choices = c(2,3,4),width=500),width=10),
               mainPanel(verbatimTextOutput("Data_Final"),width=10))),
    tabPanel("4-Association Rule(Apriori Algorithm)",
              sidebarLayout(
               sidebarPanel(textInput("Min_Support","Enter Minimum Support ( value must between 0.001:1 ):"),textInput("Min_Confidence","Enter Minimum Confid
               mainPanel(verbatimTextOutput("Data_Association"),width=10))),
    tabPanel("About Us",
              sidebarLayout(
               sidebarPanel(h4("This Project Implemented BY(نظم افكارك واجنا ف انتظارك) Team No.60 :"),width=50),
               mainPanel(h3(" * Team Leader : Mustafa Younis  -------> 23011550 "),
                         h3(" * Abdulrahman Gaber -------> 23011312"                ),
                         h3(" * Ezzat Abdel Ghaffar-------> 23011360"                 ),
                         h3(" * kirolos Mounir -------> 23011424 " ),
                         h3(" * Abdulrahman Mohammed -------> 23012002"                 ),
                         h3(" * Mina Tharwat -------> 23011576 " ),
                         h4("Thanks"),width=50,height=50)))
    )
  )
```

-First, We created fluid page and selected its theme ("cyborg") and gave a titlepanel “Analytical Program From Team NO.60”, Then we used tabsetpanel was used to create a certain number of panels separated from each other.

-First Panel : its name is Getting ready , it has one textinput() function to take the location path from the user, and has two action Button ()function named (Read Data , Clean Data).

-Second Panel : its name is Dashboard that show an analytical windows , it has only one function that is plotOutput() function to display a graphs and plots from server.

-Third Panel : its name is Clustring to apply K-Means, we devided it to twice windows by using sidebarlayout() function that contains sidebarpanel()function to display first part that has selectinput()function,to take the number of clusters from user, and also contains the mainpanel()function to display the second part and this function his task to show the clustered data from the server.

-Fourth Panel: its name is Association Rule(AprioriAlgorithm), it also devided into two windows by using sidebarlayout() function , as we mentioned the first window function is called sidebarpanel() , this function has two textinput() functions to take support and confidence values from user , also its has sliderinput() function to Allows user to select minlength by easy way, then there is second part of this panel that created by mainpanel() function , this function receive the Associated data from server by using verbatimtextOut() function.

-Fifth Panel : its name is About Us , and its task to display team's members by using h() function.

-Explanation of the Server :

−In this code, we have written special codes that allow the user to enter the location (Path) of his own file that he wants to read and clean, which leads to the rest of the operations and analysis of his data Over the Four panels.

```
#Shiny_Server

server<- function(input , output ,session ){

 #First_Panel_(Read_Data_Button)
 observeEvent(input$Read_Button,{

    Data_Path<-input$Data_Path

    output$Data_Path<-renderText({
      print(paste("Your File Path: ",input$Data_Path))
    })

    if(file.exists(Data_Path)){

      Data<-read.csv(Data_Path)

      Counter_Numeric<-0
      Counter_Character<-0

      output$Data_Str<-renderPrint({
        print(paste("Here We Have",length(Data),"Columns (Variables) , and ",nrow(Data),"Rows. 😊"))

        for(x in 1:ncol(Data)){
         if(is.character(Data[,x])){
          Counter_Character<-Counter_Character+1
         }else if(is.numeric(Data[,x])||is.integer(Data[,x])){
            Counter_Numeric<-Counter_Numeric+1
          }
        }

        cat("There is:
          (",Counter_Numeric,")Rows As Numeric Data Type
          (",Counter_Character,")Rows As Charcter Data Type\n")
        cat("\n")

        for(x in 1:ncol(Data)){
          print(paste(x,"-",names(Data)[x],"Column (",class(Data[,x]),")--->",Data[1,x]))
        }
      })
    }else{
      output$Data_Str<-renderText("File not found 🕷️📄❌☹️")
    }
 })
```

-The general task of this part is when user click on the (Read Data) button , the program return entered path to the UI by (Data_Path) variable ,Then print this path in userinterface, then check if this file already exits in user PC or not , then print a full details about CSV file using cat() function , print() function and Paste()function.

```
#First_Panel_(Clean_Data_Button)
observeEvent(input$Clean_Button,{

   Data_Path<-input$Data_Path

   if(file.exists(Data_Path)){

     Data<-read.csv(Data_Path)

     output$Data_Clean <- renderPrint({

      #Remove_Duplicated_Rows
       if(sum(duplicated(Data))!=0){
         print(paste("number of Duplicated Rows : ",sum(duplicated(Data))))
         Data<-unique(Data)
         print("Duplicated Rows Has Been Deleted♻️")
       }else{
         print("There isn't Duplicated Rows👏🟩")
       }
       print("-------------------------------------------------------------------------------------")

      #Remove_NA_Values
       if(sum(is.na(Data))!=0){
         print(paste("number of NA Values : ",sum(is.na(Data))))
         Data<-na.omit(Data)
         print("NA Value's Rows Has Been Deleted♻️")
       }else{
         print("There isn't NA Values👏🟩")
       }
       print("-------------------------------------------------------------------------------------")

      #Remove_Outliers
       Data_Edit<- subset(Data, select = -which(sapply(Data, is.character)))

       for(x in 1:ncol(Data_Edit)) {
         #كوندهن عشان يتماشى مع الفايل اللي عندنا وميمسحش الاوتلايرز بتوع عمود الكونت
         if(max(Data_Edit[,x]) - min(Data_Edit[,x]) < 50) {
           print(paste("There isn't any outliers in ", names(Data_Edit)[x], "column👏🟩"))
           print("-----------------")
            next
         }

         outlier <- boxplot(Data_Edit[,x], plot = FALSE)$out
         if(length(outlier) != 0) {
           print(paste(names(Data_Edit)[x], "column has", length(outlier), "outliers:", outlier))
           Data_Edit[-which(Data_Edit[,x]%in%outlier),]
           print("Outliers Has Been Deleted♻️")
         }else {
           print(paste("There isn't any outliers in ", names(Data_Edit)[x], "column👏🟩"))
         }
         print("-----------------")
       }


     })
   }else{
     output$Data_Clean<-renderText("File not found 🕷️🟩🟥")
   }
```

-this part task is when user click in clean Button , the Data Cleaning process performed , but after check existing file in user's device , then program implement the main points of cleaning data of CSV file As follows:

1- Remove Duplicated Rows using Unique() function.

2- Delete outlires using

Data_Edit[which(Data_Edit[,x]%in%outlier),]

3- Delete NA values using na.omit() function.

```
#Second_Panel_(Dashboard)
output$Dashboard <- renderPlot({

   Data_Path<-input$Data_Path
   par(mfrow=c(2,2))
   Data<-read.csv(Data_Path)
   Data<-unique(Data)


   #Distribution_Of_Total_Spending_(Box_Plot)
   boxplot(Data$total,col="skyblue",ylab="Total Spending",main="Distribution Of Total Spending",width=450)

   #City_Vs_Total_Spending_(Bar_Plot)
   City_total<-Data %>%
   group_by(city) %>%
   summarize(total=sum(total))
   City_total <- arrange(City_total, desc(total))
   n <- length(unique(City_total$city))
   city_colors <- brewer.pal(n, "Set2")
   barplot(height =City_total$total,
           pch = 19,
           las = 2,
           cex.names = 1.2,
           name=City_total$age,
           names.arg = City_total$city,
           main="Total Spending by each City",
           col=city_colors,
           ylim=c(0,3000000))
   legend("topright", legend = unique(City_total$city), col = city_colors,title="City" ,pch = 19, bty = "n")


   #Cash_Vs_Credit_(Pie_Chart)
   t<-table(Data$paymentType)
   percentage<-paste0(round(100*t/sum(t),digits=2),"%")
   pie(t , labels=percentage , main="Compare cash and credit totals", col=c("lightblue","pink",width=500),cex = 1.5,init.angle = 90)
   legend("bottomleft",title="Payment Type",legend = c("cash","credit"),fill = c("lightblue","pink"))


   #Age_Vs_Total_Spending_(Bar_Plot)
   summary_age_total<-Data %>%
   group_by(age) %>%
   summarize(total=sum(total))
   barplot(height = summary_age_total$total,
           name = summary_age_total$age,
           xlab = "age",
           ylab = "total",
           main = "total spending by age",
           col="orange",
           ylim=c(0,2000000))

},bg = "whitesmoke")
```

-this part task it to take the four graphs and put them in one window using par(mfrow=c(2,2)) , then send this graphs to UI using renderplot() function.

```
#Third_Panel_Kmeans_(Clustring)
observe({

 if(file.exists(input$Data_Path)){
     Data_Path<-input$Data_Path
     Data<-read.csv(Data_Path)
     Data<-unique(Data)
     Data<-Data[,-c(1,2,7,8)]

     Total_Vector<-rep(0,max(Data$rnd))
     Name_Vector<-rep(NA,max(Data$rnd))
     Age_Vector<-rep(0,max(Data$rnd))


     for(i in 1:max(Data$rnd)){
      for(x in 1:nrow(Data)){
       if(i == Data[x,2]){
         Total_Vector[i]<-Total_Vector[i]+Data[x,1]
         Name_Vector[i]<-Data[x,3]
         Age_Vector[i]<-Data[x,4]
       }else{
         next
         }
      }
     }

   #Verify
   sum(Data$total)-sum(Total_Vector)

   Data_K<-data.frame(Name_Vector,Total_Vector,Age_Vector)
   names(Data_K)<-c("Name","Total","Age")
   Data_Kk<-Data_K[,-1]

   Data_Kmeans<-kmeans(Data_Kk,centers=input$Number_Of_Clusters)

   Cluster<-Data_Kmeans$cluster

   Data_Final<-cbind(Data_K,Cluster)

   output$Data_Final<-renderPrint({print(Data_Final)})

   }else{
     output$Data_Final<-renderPrint({print("File not Found 🔍📂❌")})
 }

})
```

This part task is to perform K-Means Clustring using kmeans() after omit character columns and check this csv file exist in user device , then send clustered Data to UI ,finally displays each customer name , age , Total Spending , his Cluster, in general it creates a k-means

algorithm with the number of clusters that the user chooses, from two to four. We have designed the user interfaces so that they allow him to choose between them without the need to write, and also create a table in which the data is displayed without repetition, and it also displays the number of clusters.

```
#Fourth_Panel_Association_Rule_(Apriori)
observe({

   if(file.exists(input$Data_Path)){

    Data_Path<-input$Data_Path
    Data<-read.csv(Data_Path)
    unique(Data)

     if(0.001 <= input$Min_Support && input$Min_Support <= 1 &&
        0.001 <= input$Min_Confidence && input$Min_Confidence <= 1 &&
        2 <= input$Min_Length && input$Min_Length <= 10){

          Items_Excel<-paste(Data$items)
          write(Items_Excel , file = "Items.txt")
          Data <- read.transactions("Items.txt",sep=",")
          Apriori<-apriori(Data,parameter=list(supp=as.numeric(input$Min_Support),conf=as.numeric(input$Min_Confidence),minlen=as.numeric(input$Min_Length)))

          output$Data_Association<-renderPrint({
           print("Right Values,Here We Go: 🎉")
           print(Apriori)
           options(max.print = 9999)
           inspect(Apriori)
          })

     }else{ output$Data_Association<-renderPrint({print("Values Isn't Stable 😐")})}

   }else{output$Data_Association<-renderText({print("File not Found 🕵️‍♂️")})}
})
```

This part role is to perform Apriori algorithm using Apriori() function , after check this csv file exist in user device and convert items column to txt file, in general program creates an association rule algorithm that allows the user to choose the number of Support & Confidance from .001 to 1 when making the roll and then display the result on the user interface that we explained previously And display the products in a table with all the data for each product.

-Explanation of the ShinyApp :

```
#Shiny_App
shinyApp(ui = ui , server = server)
#Done
```

It is with the last line of code(ShinyApp) that allows this entire program to run.

Thanks ❤ ❤ ❤ ❤ ❤ ❤ ❤ ❤ ❤